GenAI Used: Gemini (accessed on KU Linux machine using Firefox)

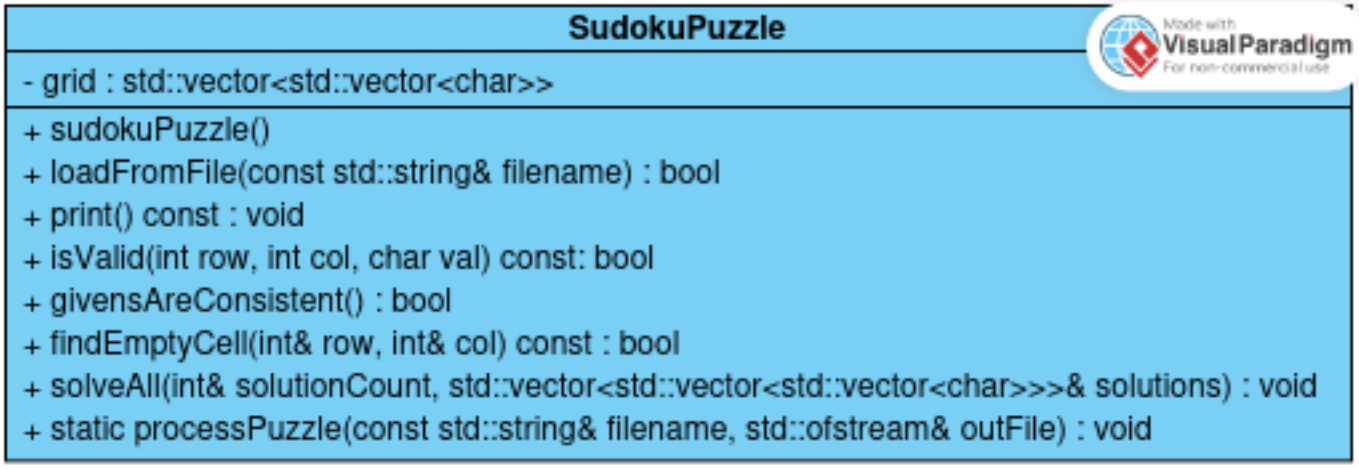
Prompt Given:

{Assignment6.cpp included as attachment}

Attached is a C++ program that solves sudoku puzzles from text files. When the program solves a puzzle, it writes the results of all the puzzles to one text file.

Given the program, create an outline for a UML Class Diagram. Each class should have its attributes, methods, and relationships between other classes listed. The attributes for each class should have their visibility, name, and type listed. The methods for each class should have their visibility, name, parameters (and their types), and return types listed.

Graphic made with Visual Paradigm using GenAI output as a guide:



Class: SudokuPuzzle

Attributes:

- grid : std::vector<std::vector<char>>

Methods:

+ sudokuPuzzle()
+ loadFromFile(const std::string& filename) : bool
+ print() const : void
+ isValid(int row, int col, char val) const: bool
+ givensAreConsistent() : bool
+ findEmptyCell(int& row, int& col) const : bool
+ solveAll(int& solutionCount, std::vector<std::vector<std::vector<char>>>& solutions) : void
+ static processPuzzle(const std::string& filename, std::ofstream& outFile) : void